



Documentation of Expense Management System

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Expense Manager - Complete Project Documentation

1. Project Introduction

This Expense Manager is a full-stack web application developed using React for frontend, Node.js & Express for backend, and MongoDB as database. It allows users to manage their income and expenses, track financial data, and visualize analytics using charts.

2. Project Features

- User Signup & Login Authentication
- Secure Password Handling
- Add / Edit / Delete Transactions
- Filter Transactions (Daily, Monthly, Yearly)
- Search by Title
- Category Filtering
- Date Range Filtering
- Dashboard Summary (Income, Expense, Balance)
- Budget Limit Alert System
- Dark Mode Toggle
- Analytics Page (Charts)
- Highest Spending Category Detection
- Savings & Savings Rate Calculation

3. Project Flow

1. User opens Landing Page
2. User goes to Signup/Login
3. After login → Dashboard opens
4. User adds transactions
5. Data stored in MongoDB via backend API
6. Dashboard shows summary + history
7. User can filter/search data
8. Analytics page shows charts & insights

4. Folder Structure

Backend:

- models (User, Transaction)
- controllers (authController, transactionController)
- routes (authRoutes, transactionRoutes)
- middleware (authMiddleware)
- server.js

Frontend:

- src/pages (Login, Signup, Dashboard, Analysis, Landing)
- src/components
- api.js (API calls)
- App.jsx (routing)

5. Backend Explanation

Models: Define database schema (User, Transaction)

Controllers: Handle logic like signup, login, CRUD operations

Routes: Define API endpoints

Middleware: Used for authentication (JWT)

Server.js: Main backend entry point

6. Frontend Explanation

Landing Page: Navigation (Login/Signup)

Login Page: User authentication

Signup Page: Create account with validation

Dashboard: Manage transactions, filters, dark mode

Analysis Page: Charts (Line, Bar, Pie)

7. REST APIs Used

Auth APIs:

POST /api/auth/signup

POST /api/auth/login

Transaction APIs:

GET /api/transactions

POST /api/transactions

PUT /api/transactions/:id

DELETE /api/transactions/:id

8. Installation Commands (VS Code)

Backend:

npm init -y

npm install express mongoose cors dotenv

Frontend:

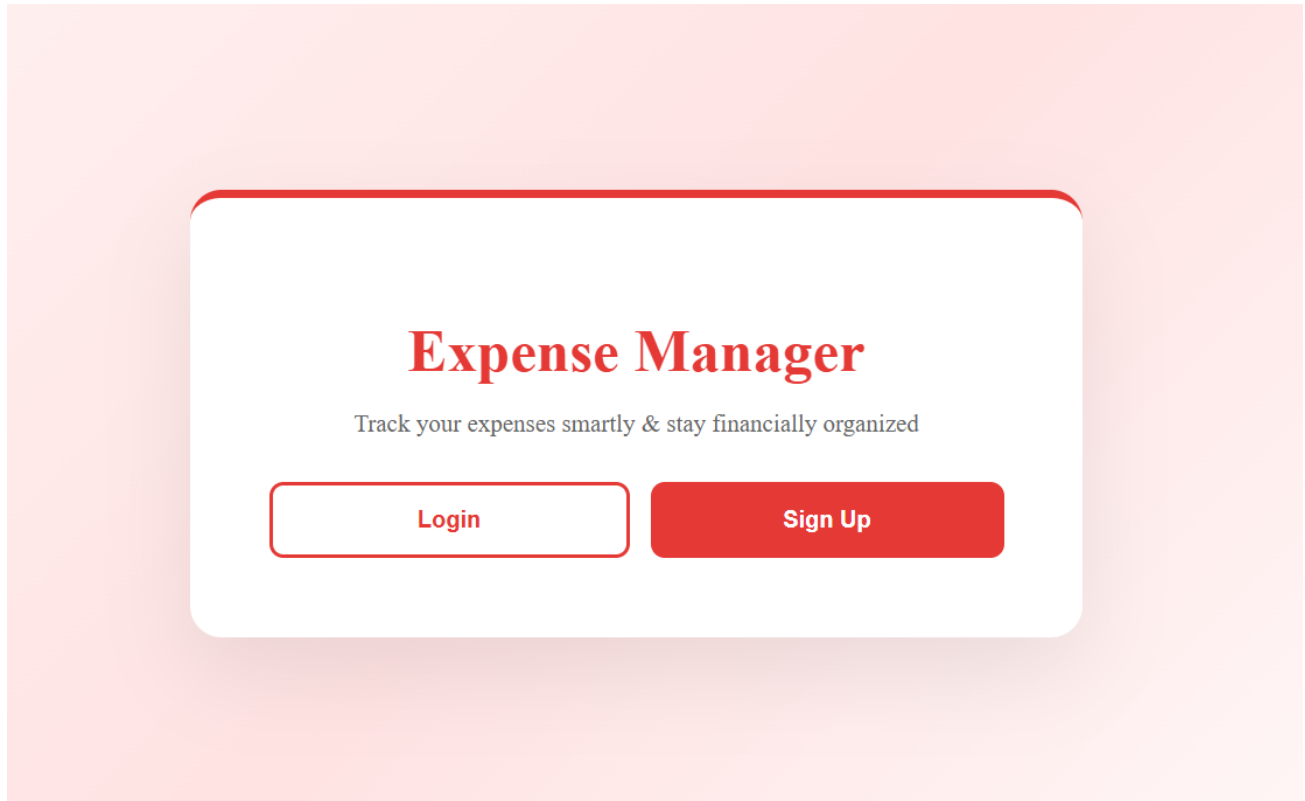
npm create vite@latest

```
npm install  
npm install recharts react-router-dom
```

9. Conclusion

This project demonstrates full-stack development with authentication, CRUD operations, filtering, and data visualization. It helps users manage finances efficiently.

Screenshots:



Login

[Show](#)

Login

Forgot Password? [Reset Here](#)

Don't have account? [Sign Up](#)

Sign Up

[Show](#)[Show](#)

Create Account

Already have account? [Login](#)

Forgot Password



Change Password

[Back to Login](#)

Transaction Manager



 Balance: 193500

↑ Income: 250000

↓ Expense: 56500

 Warning: Budget Exceeded!

Add Transaction



ALL

DAILY

MONTHLY

YEARLY

History (6)

30/03/2026

Add Transaction

Breakfast - 500
Food | 2026-03-30

EditDel

Walk - 1000
horse riding | 2026-03-31

EditDel

salary - 40000
marketing | 2026-03-22

EditDel

hotling - 10000
food | 2026-01-28

EditDel

salary - 250000
freelance | 2026-03-29

EditDel

shopping - 5000
clothes | 2027-08-29

EditDel

Set Budget Limit: 10000

hotling - 10000
food | 2026-01-28

EditDel

salary - 250000
freelance | 2026-03-29

EditDel

shopping - 5000
clothes | 2027-08-29

EditDel

Set Budget Limit: 10000

Analytics

Logout

[← Back](#)

Financial Analytics

ALL

DAILY

MONTHLY

YEARLY

Total Savings

Rs. 193500

77.4% of Income saved

Top Expense Category

marketing

Highest Spending

Total Transactions

6

In selected period

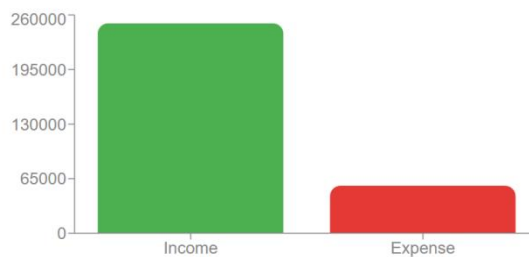
Income vs Expense Trend



Budget Distribution



Income vs Expense Bar

[← Back](#)

Financial Analytics

ALL

DAILY

MONTHLY

YEARLY

Total Savings

Rs. -500

0% of Income saved

Top Expense Category

Food

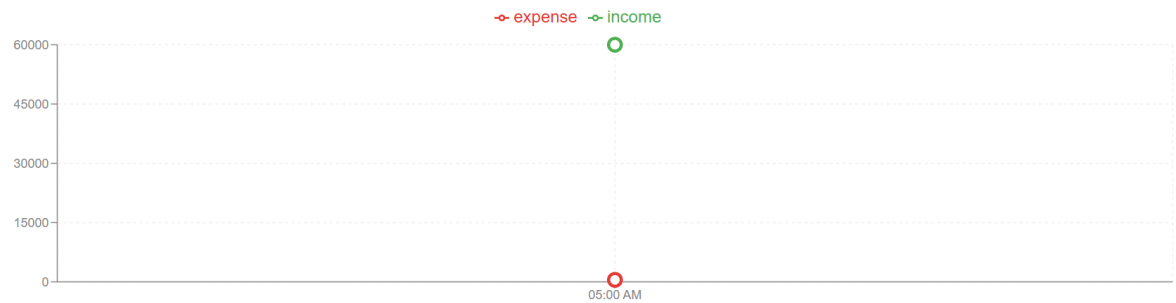
Highest Spending

Total Transactions

1

In selected period

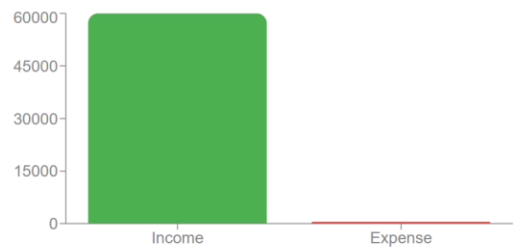
Income vs Expense Trend



Budget Distribution



Income vs Expense Bar



← Back

Financial Analytics

ALL

DAILY

MONTHLY

YEARLY

Total Savings

Rs. 268500

86.6% of income saved

Top Expense Category

marketing

Highest Spending

Total Transactions

6

In selected period

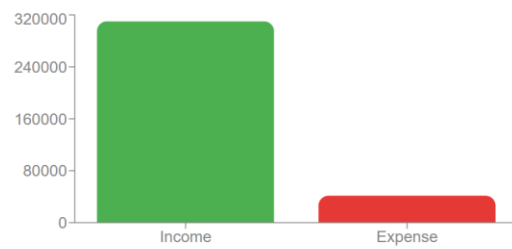
Income vs Expense Trend



Budget Distribution



Income vs Expense Bar



[← Back](#)

Financial Analytics

ALL

DAILY

MONTHLY

YEARLY

Total Savings

Rs. 258500

83.4% of income saved.

Top Expense Category

marketing

Highest Spending

Total Transactions

7

In selected period.

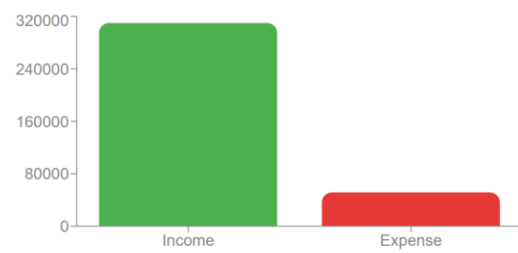
Income vs Expense Trend



Budget Distribution



Income vs Expense Bar



Recording:



Recording Expense management System.mp4